

Science Lab Report: Cup Material and Heat Retention in Water

1. Investigation Question

Which type of cup material keeps hot water warm for the longest time?

2. Independent Variable

Type of cup material (Paper, Glass, Ceramic, and Foam).

3. Dependent Variable

Water temperature after 15 minutes (measured in degrees Celsius, °C).

4. Control Variables / Group

Control Variables: Initial water temperature (80°C), water volume (200 mL), room temperature (22°C), cup volume (250 mL), and no lids on any cups. Control Group: Paper cup.

5. Hypothesis

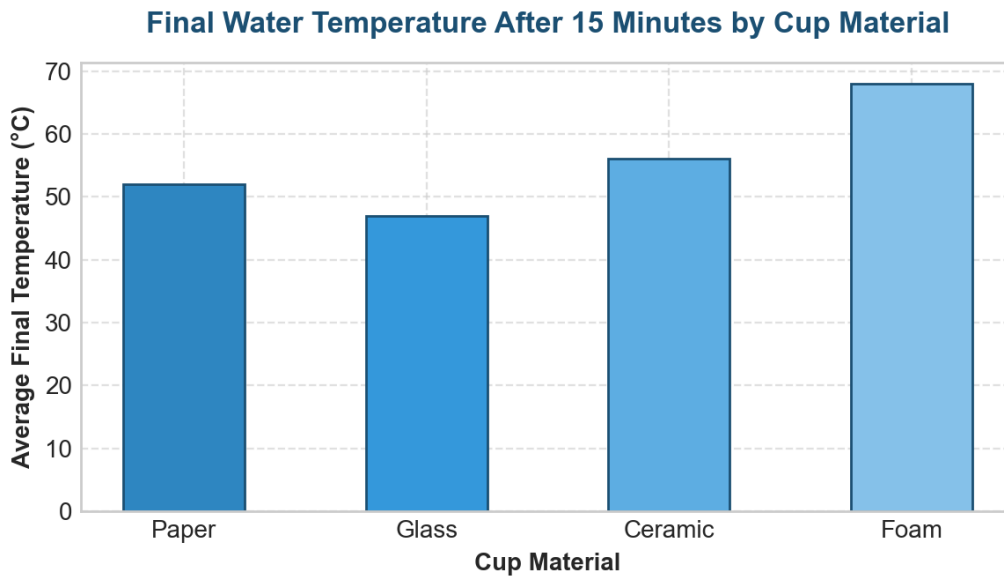
If a foam cup is used, then the final water temperature will be higher (in degrees Celsius, °C) after 15 minutes than with paper, glass, or ceramic cups because foam is a better thermal insulator.

6. Experimental Procedure

1. Place four clean cups (Paper, Glass, Ceramic, Foam) on a level table.
2. Pour 200 mL of hot water at exactly 80°C into each cup.
3. Start a timer for 15 minutes.
4. Keep the room temperature steady at 22°C without turning on any fans.
5. At 15 minutes, insert a digital thermometer into the center of each cup and record the final temperature in °C.
6. Repeat the experiment 3 times to get reliable averages.

7. Results

Cup Material	Trial 1 (°C)	Trial 2 (°C)	Trial 3 (°C)	Average Final Temp (°C)
Paper	52.0	51.5	52.5	52.0
Glass	47.0	46.5	47.5	47.0
Ceramic	56.0	55.0	57.0	56.0
Foam	68.0	67.5	68.5	68.0



8. Analysis

The foam cup retained heat best, keeping the water at an average of 68.0°C. Ceramic was second (56.0°C), paper was third (52.0°C), and glass allowed the most heat loss, finishing at 47.0°C. Foam kept the water 21.0°C warmer than glass.

9. Conclusion

The hypothesis was supported. If a foam cup is used, then the final water temperature is higher (in °C) after 15 minutes. Foam contains tiny trapped air pockets that prevent heat from escaping through conduction.

Science Lab Report: Ball Mass and Sand Crater Depth

1. Investigation Question

How does the mass of a dropped ball affect the depth of the crater it creates in sand?

2. Independent Variable

Mass of the ball (measured in grams, g). We tested 20 g, 50 g, 100 g, and 150 g balls.

3. Dependent Variable

Depth of the crater created in the sand (measured in millimetres, mm).

4. Control Variables / Group

Control Variables: Drop height (1.0 metre), ball diameter (same 4 cm size for all balls), type and dryness of sand, and smoothing the sand flat before each drop. Control Group: The lightest ball (20 g).

5. Hypothesis

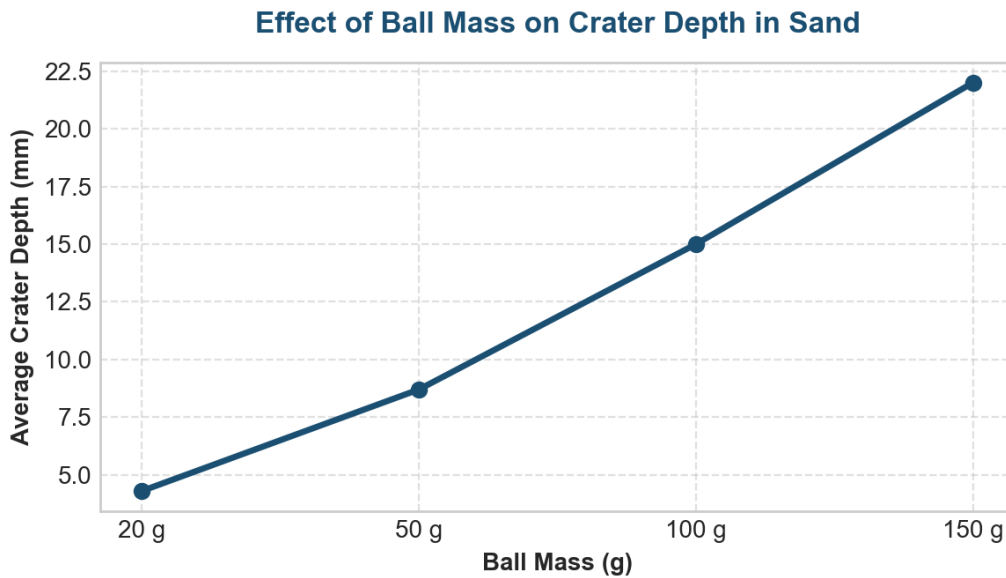
If the mass of the ball is increased, then the depth of the crater in the sand will increase (in millimetres, mm) because a heavier ball hits the sand with more kinetic energy.

6. Experimental Procedure

1. Fill a plastic tray with dry sand to a depth of 10 cm and smooth the top flat with a ruler.
2. Hold the 20 g ball at a height of 1.0 metre above the sand surface.
3. Drop the ball gently into the sand without throwing it.
4. Carefully lift the ball out of the sand without disturbing the crater.
5. Insert a thin ruler to measure the depth of the crater in millimetres (mm).
6. Smooth the sand and repeat 3 times for each ball mass (20 g, 50 g, 100 g, 150 g).

7. Results

Ball Mass (g)	Trial 1 (mm)	Trial 2 (mm)	Trial 3 (mm)	Average Depth (mm)
20 g	4	5	4	4.3
50 g	9	8	9	8.7
100 g	15	16	14	15.0
150 g	22	21	23	22.0



8. Analysis

The data shows that heavier balls made significantly deeper craters. The 20 g ball produced a crater of only 4.3 mm, while the 150 g ball created a crater 22.0 mm deep. Every increase in ball mass resulted in a deeper impact mark.

9. Conclusion

The hypothesis was supported. If the mass of the ball is increased, then the crater depth increases (in mm). When dropped from the same height, an object with greater mass delivers more force upon impact, displacing more sand.

Science Lab Report: Parachute Canopy Size and Fall Time

1. Investigation Question

How does the canopy size of a parachute affect the time it takes to drop from a height of 2 metres?

2. Independent Variable

Canopy diameter/size (measured in centimetres, cm). We tested square plastic canopies measuring 15 cm, 25 cm, 35 cm, and 45 cm on each side.

3. Dependent Variable

Time taken for the parachute to hit the floor from 2 metres high (measured in seconds, s).

4. Control Variables / Group

Control Variables: Drop height (2.0 m), string length (20 cm), payload mass (1 metal paperclip, 5 g), and no wind in the room. Control Group: The smallest canopy size (15 cm).

5. Hypothesis

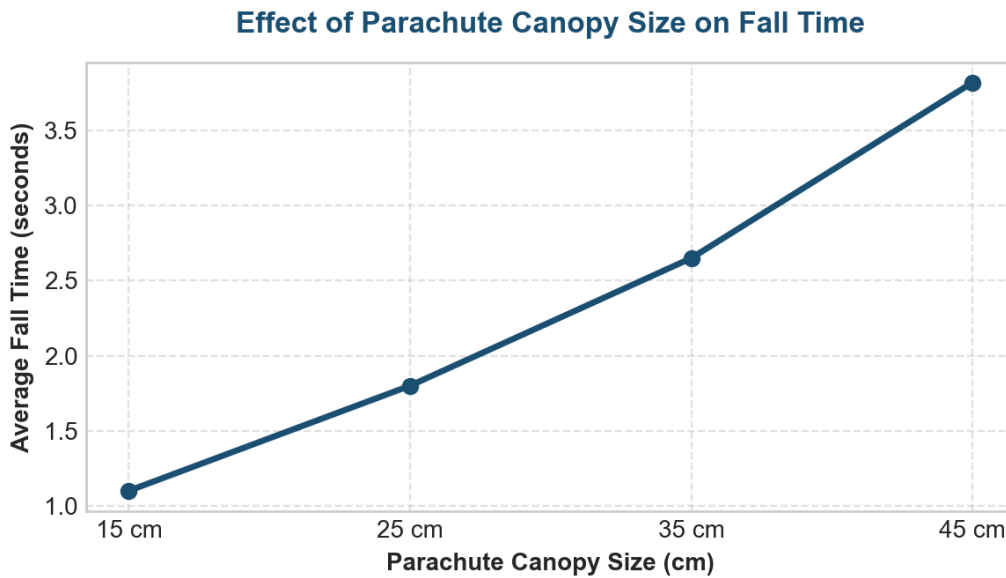
If the parachute canopy size is increased, then the parachute will take more time (in seconds) to reach the ground because a larger surface area creates more air resistance.

6. Experimental Procedure

1. Cut plastic sheets into four squares: 15x15 cm, 25x25 cm, 35x35 cm, and 45x45 cm.
2. Attach four 20 cm strings to the corners of the 15 cm canopy and tie them to a 5 g paperclip.
3. Hold the top of the parachute at a marked height of 2.0 metres.
4. Drop the parachute and start the stopwatch at the exact same moment.
5. Stop the stopwatch when the paperclip touches the floor. Record the time in seconds.
6. Perform 3 trials for each parachute size.

7. Results

Canopy Size (cm)	Trial 1 (s)	Trial 2 (s)	Trial 3 (s)	Average Time (seconds)
15 cm	1.10	1.05	1.15	1.10
25 cm	1.80	1.75	1.85	1.80
35 cm	2.65	2.70	2.60	2.65
45 cm	3.80	3.90	3.75	3.82



8. Analysis

The data demonstrates that larger parachutes fell at slower speeds. The 15 cm parachute dropped quickly in 1.10 seconds, while the 45 cm parachute stayed in the air for an average of 3.82 seconds. The extra surface area increased the fall time by 2.72 seconds.

9. Conclusion

The hypothesis was supported. If the parachute canopy size is increased, then the fall time (in seconds) increases. A larger surface area captures more air molecules underneath, creating greater upward drag (air resistance) that slows the fall.

Science Lab Report: Water Temperature and Sugar Dissolving Speed

1. Investigation Question

How does the temperature of water affect the time it takes for sugar to dissolve completely?

2. Independent Variable

Water temperature (measured in degrees Celsius, °C). We tested 20°C, 40°C, 60°C, and 80°C.

3. Dependent Variable

Time taken for 10 grams of sugar to dissolve completely (measured in seconds, s).

4. Control Variables / Group

Control Variables: Volume of water (150 mL), type of sugar (white table sugar), mass of sugar (10 g), and stirring speed (stirred 2 times per second). Control Group: Room temperature water at 20°C.

5. Hypothesis

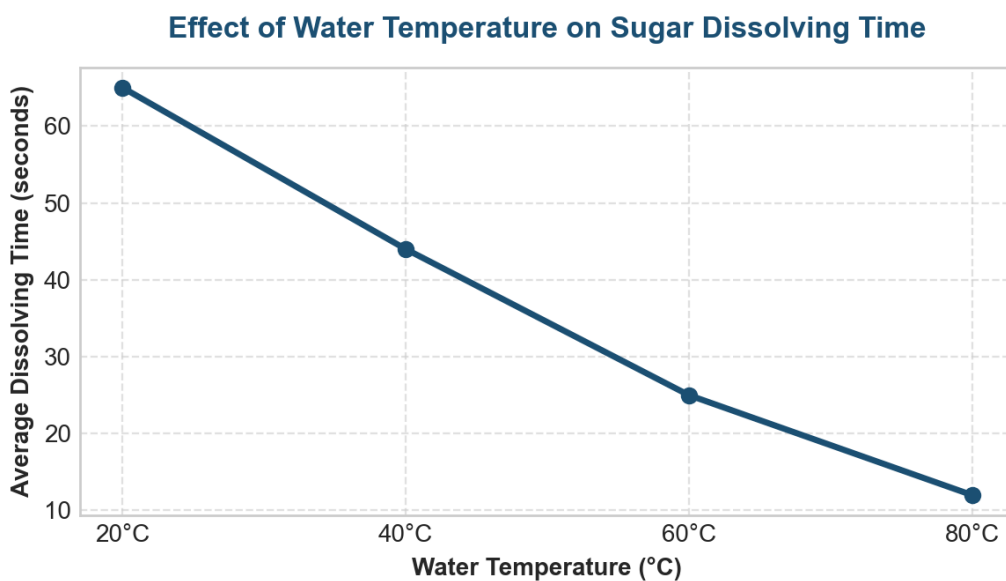
If the water temperature increases, then the time it takes for the sugar to dissolve will decrease (in seconds) because hot water particles move faster and break down the sugar crystals quickly.

6. Experimental Procedure

1. Measure 150 mL of water and heat it to 20°C in a clear glass beaker.
2. Add exactly 10 grams of white sugar to the water.
3. Start the stopwatch immediately and stir at a steady speed.
4. Stop the stopwatch as soon as all sugar grains disappear and record the time in seconds.
5. Repeat the test 3 times for 20°C.
6. Repeat steps 1 to 5 using water heated to 40°C, 60°C, and 80°C.

7. Results

Water Temp (°C)	Trial 1 (s)	Trial 2 (s)	Trial 3 (s)	Average Time (seconds)
20°C	65	63	67	65.0
40°C	44	46	42	44.0
60°C	26	24	25	25.0
80°C	12	11	13	12.0



8. Analysis

The data shows a clear pattern: as water temperature increases, dissolving time decreases. At 20°C, it took an average of 65.0 seconds for sugar to dissolve. At 80°C, it took only 12.0 seconds. The sugar dissolved 53.0 seconds faster in the hottest water compared to the coldest water.

9. Conclusion

The hypothesis was supported. If the water temperature increases, then the sugar dissolves in fewer seconds. This happens because higher temperatures give water molecules more kinetic energy to break apart sugar particles faster.

Science Lab Report: Ramp Height and Toy Car Travel Distance

1. Investigation Question

How does the height of a ramp affect the distance a toy car travels across a flat floor?

2. Independent Variable

Height of the ramp (measured in centimetres, cm). We tested heights of 5 cm, 10 cm, 15 cm, and 20 cm.

3. Dependent Variable

Distance traveled by the toy car across the floor (measured in centimetres, cm).

4. Control Variables / Group

Control Variables: Same toy car (mass of 45 g), same ramp surface (smooth wood), same floor surface (tile), and releasing the car without pushing it. Control Group: The lowest ramp height of 5 cm.

5. Hypothesis

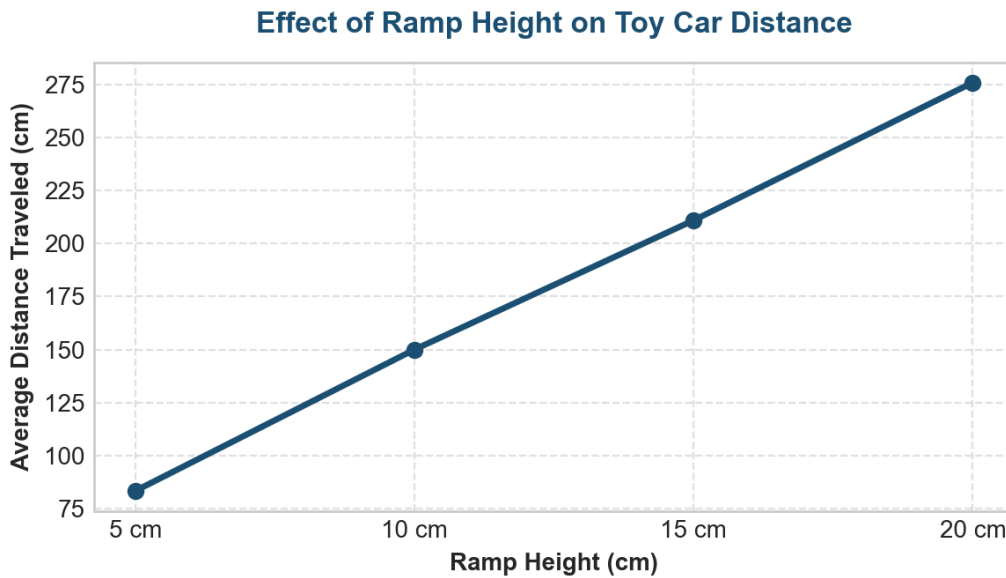
If the ramp height is increased, then the toy car will travel a greater distance (in centimetres) because higher ramps give the car more potential energy to turn into motion.

6. Experimental Procedure

1. Set the ramp height to 5 cm using wooden blocks.
2. Place the front wheels of the toy car at the top edge of the ramp.
3. Release the car gently without pushing it.
4. Use a measuring tape to measure the distance from the bottom of the ramp to the front of the car in centimetres (cm).
5. Repeat the test 3 times for 5 cm.
6. Increase the ramp height to 10 cm, 15 cm, and 20 cm, and repeat steps 2 to 5 for each height.

7. Results

Ramp Height (cm)	Trial 1 (cm)	Trial 2 (cm)	Trial 3 (cm)	Average Distance (cm)
5 cm	82	85	83	83.3
10 cm	148	152	150	150.0
15 cm	210	215	208	211.0
20 cm	275	280	273	276.0



8. Analysis

The results indicate that increasing ramp height increases the distance traveled by the toy car. At a height of 5 cm, the car rolled an average of 83.3 cm. When the height was raised to 20 cm, the average distance increased to 276.0 cm, which is an increase of 192.7 cm.

9. Conclusion

The hypothesis was supported. If the ramp height is increased, then the toy car travels a longer distance (cm). This happens because a higher starting position provides more gravitational potential energy, which converts into kinetic energy.